MIL-V-15020F(SHIPS)
16 October 1968
SUPERSEDING
MIL-V-0015020E(SHIPS)
1 December 1966
MIL-V-15020D
10 July 1962
(See 6.3)

MILITARY SPECIFICATION

VALVES, FLUSH (SHIPBOARD)

1. SCOPE

1.1 Scope. This specification covers flush valves with flow-control stop valves for shipboard use.

1.2 Classification. Flush valves shall be of the following types and classes, as specified (see 3.2.2, 3.2.3 and 6.2):

Type I - Water closet flush valve, 1 inch inlet, 1-1/2 inch outlet.
  Class A - Outlet connection for shock mounted installations.
  Class B - Outlet connection for rigid mounted installations.
Type II - Urinal flush valve, 1/2 inch inlet, 3/4 inch outlet.
  Class A - Outlet connection for shock mounted installations.
  Class B - Outlet connection for rigid mounted installations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-N-290 - Nickel Plating (Electrodeposited).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-P-116 - Preservation, Methods Of.
MIL-R-880 - Rubber, Medium-Soft (For Airport, Airtight, and Watertight Door, and Hatch Gaskets).
MIL-G-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-B-15395 - Brazing Alloys, Silver.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 61-63 Steam or Valve Bronze Castings.
B 62-63 Composition Bronze or Ounce Metal Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Materials used in the construction of the flush and control stop valves shall be as specified in 3.1.1 through 3.1.5. Where valves have parts which are not covered herein or are not exposed to flush water, materials shall be of the best commercial grade.

3.1.1 Valve bodies. Valve bodies, bonnets and packing nuts shall be of cast bronze in accordance with ASTM B61 or B62.

3.1.2 Flow control valve internal parts. Where exposed to flush water, the stem of the flow control valve shall be nickel-copper alloy. The disc shall be nickel-copper alloy or nonmetallic material as approved by the procuring activity. The seat may be integral with the body.

3.1.3 Flush valve internal parts. The internal parts of the flush valve shall be of the best commercial material suitable for a marine environment. Brass shall not be used for parts exposed to flush water.

3.1.4 Rubber parts. Rubber parts, such as valve discs, sleeves, or slip joint washers, shall be of material conforming to MIL-R-880, MIL-G-1149, or as approved by the procuring activity.

3.1.5 Outlet tubing. Outlet tubing shall be of high copper content alloy.

3.1.6 Brazing ring. The material of the brazing ring shall be in accordance with MIL-B-15395.

3.2 Design. Types I and II flush valves shall be of a sealed design, suitable for use in sea water, or a contaminated harbor and river water environment. The design shall be such that water does not contact the control mechanism, and is not required to actuate the valve. Valve body assemblies shall be identical for each type of valve.

3.2.1 Flush valves shall have a 1-inch angle flow control stop valve connected to the flush valve inlet by a 1-inch union joint. The flow control stop valve shall have a screw driver slotted stem and lock nut or protective cap. The inlet to the flow control stop valve shall be in accordance with figure 1. When required, an inlet adapter with insert silver brazing ring in accordance with figure 2 shall be furnished for the flow control valve (see 6.2). Inlet adapter material shall be in accordance with MIL-C-15726 or ASTM B61.

3.2.1.1 Unless otherwise specified (see 6.2), the distance between the centerline of the flush valve and the centerline of the angle flow control stop valve shall be 4-3/4 inches.

3.2.2 Class A - outlet connections for shock mounted installations (see 6.2).

3.2.2.1 The outlet connections of the type I flush valve shall consist of a 5 inch length of 1-1/2 inch outside diameter (o.d.) tubing, coupling nut, slip joint packing and friction ring (see figure 3). A flared connection as shown on figure 4 is acceptable in lieu of a slip joint.

3.2.2.2 The outlet connections of the type II flush valve shall consist of a 2-1/4 inch long reducing fitting with a 1/2 inch offset, brazed to a 4-1/8 inch length of 3/4 inch tubing (see figure 4). Coupling nut and rubber washer shall be included.

3.2.2.3 Outlet connection tubing shall be scored for 1-1/2 inches (to prevent hose slippage) prior to plating.

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3.2.3 Class B - outlet connections for rigid mounted installations (see 6.2).

3.2.3.1 The connection of the type I flush valve shall be provided with an outlet tube; spud coupling nut, spud flange, slip joint packing and friction ring (see figure 5).

3.2.3.2 The connection of the type II flush valve shall have an adapter piece, valve outlet tube, adapter nut, rubber washer and friction washer (see figure 6).

3.2.4 Operation. The flush valves shall be capable of continuous reliable operation without sticking, chattering, or causing water hammer. Means for regulating the flushing time shall be provided within the valve. Adjustments shall be made without shutting off the water or removing the valve from the line.

3.2.5 Valve seats. There shall be no metal-to-metal contact between the valve disc and its seat. Valve discs shall be rubber (see 3.1.4) and shall be easily renewable.

3.2.6 The outlet tube union nut packing shall be backed up with a brass or polyethylene antifriction washer for protection against damage when tightening the union nut.

3.2.7 Application of springs. Springs shall not be used where exposed to sea water. Spring used in closing the valve shall be protected from sea water by a rubber boot or sleeve.

3.2.8 Threads. All threads shall conform to Handbook H28.

3.2.9 Capacity.

3.2.9.1 Type I flush valves shall be capable of adjustment to control the volume of flushing water to 3 to 5 gallons of water per flush at any inlet pressure from 10 to 50 pounds per square inch (p.s.i.).

3.2.9.2 Type II flush valves shall be adjustable to deliver between 3/4 and 1-1/2 gallons of water per flush at any inlet pressure from 10 to 50 p.s.i.

3.3 Finish. Valves shall be polished and nickel-chromium plated.

3.3.1 Prior to plating, valves shall be thoroughly cleaned of sand and scale and shall have all fins and roughness removed.

3.3.2 Nickel-chromium plating shall be applied to surfaces normally visible after installation including exposed bolts, nuts, screws, or other fastenings.

3.3.3 Nickel plating on copper base alloys shall be class 1, type I, in accordance with QQ-N-290.

3.3.4 Chromium plating over nickel shall be class 1, type I, in accordance with QQ-C-320.

3.4 Drawings. A sectional assembly drawing of the valves, showing roughing in dimensions and a list of material, shall be furnished to the procuring activity for approval.

3.5 Repair parts. Each valve shall be furnished with one set of repair parts as follows:

(a) Valve seats or discs: 100 percent.
(b) Slip joint packing and friction washers: 100 percent.
(c) Rubber sleeves or boots: 100 percent.
(d) Springs: 20 percent of each type and size.

3.6 Marking.

3.6.1 Each valve shall be permanently stamped before plating with the trade mark or name of the manufacturer.

3.6.2 All outlet connection tubes shall bear the following instruction attached to the tube by suitable adhesive or adhesive tape:

"CUT TO PROPER LENGTH AT
INSTALLATION. ALLOW A MINIMUM
OF 5 INCHES BETWEEN TUBE ENDS
WHEN USING RUBBER HOSE FOR SHOCK
MOUNTED EQUIPMENT."

3.7 Workmanship. The valves shall be free from any defects which may affect the serviceability or operation. The quality of workmanship shall be first class in every respect.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all valves of the same type and class offered for delivery at the same time.

4.2.2 Sampling for examination. A random sample of valves shall be selected from each lot offered for examination in accordance with MIL-STD-105 at inspection level I. The Acceptable Quality Level shall be equal to 1.5 percent defective.

4.2.3 Sampling for hydrostatic and leakage tests. A random sample of valves shall be selected from each lot for tests specified in 4.3.2 in accordance with MIL-STD-105 at inspection level I; however the smallest sample size shall be five. The Acceptable Quality Level shall be equal to 2.5 percent defective.

4.3 Quality conformance inspection.

4.3.1 Examination. Each of the sample valves selected in accordance with 4.2.2 shall be examined to verify compliance with the requirements of this specification which do not require tests. Any valve in the sample containing one or more defects shall not be offered for delivery, and if the number of defective valves in any sample exceed the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

4.3.2 Tests. Each of the sample valves selected in accordance with 4.2.3 shall be tested as specified in 4.3.3 and 4.3.4.

4.3.3 Hydrostatic test. Parts subject to inlet pressure shall be tested to 100 p.s.i. pressure with valve gagged for strength, porosity, and leakage on seat.

4.3.4 Leakage test. Valves shall be adjusted to the delivery settings specified in 3.2.9 and when tested to 50 p.s.i. working pressure, shall show no leakage.

4.3.5 Operation tests. One sample valve from each lot shall be operated with the inlet pressure of 50 p.s.i. and the flow-control valve adjusted to give a 10 second flush cycle. The quantity of water passed shall be measured. Five discharges shall be measured with a sufficient waiting period between successive operations to disclose any faulty adjustment or defect or workmanship. After five discharges at 50 p.s.i. the inlet pressure shall be reduced to 10 p.s.i. and another set of five discharges shall be made with the inlet pressure reduced to 10 p.s.i. and with the flow control valve re-adjusted to give a 10 second flush cycle. At both pressures the requirements of 3.2.9 shall be met.

4.4 Examination of preparation of delivery. An examination shall be made to determine that packaging, packing and marking requirements of the applicable contract or order are complied with. Defects shall be scored in accordance with table I. The lot size shall be the number of shipping containers fully prepared for delivery, with the exceptions that containers need not be sealed or closed, nor interior containers or case liners sealed (if applicable). Examination shall be made in two phases; first, an interior examination in process of packaging; and second, an examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be one container prepared for delivery as set forth above. The inspection level shall be S-2 of MIL-STD-105, with an acceptable quality level (AQL) of 4.0 defects per 100 units.

Table I - Examination of preparation for delivery

Examine	Defects
Marking (interior package or container and exterior container as applicable)	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Component missing, damaged, defective or not as specified.
Workmanship (as applicable)	Inadequate or improper packaging or packing, such as closure of interior packages or containers, closure of caseliners or container flaps,

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Table I - Examination of preparation for delivery (Continued)

Examine	Defects
Contents (interior and exterior container as applicable) Weight (exterior container)	taping of seams, corners, and manufacturer's joint, closure of alternate containers; loose strapping or tape banding; inadequate stapling; bulging or distortion of containers. Number per container not as specified in the contract or order. Weight per container exceeds maximum specified in the contract or order.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

5.2 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.2.1 Flush valves.

5.2.1.1 Packaging. Packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.2.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2.2 Onboard repair parts.

5.2.2.1 Packaging. Onboard repair parts shall be packaged in accordance with method III of MIL-P-116. Unit and intermediate containers shall conform to PPP-B-566, PPP-B-676 or PPP-B-636. Parts shall be packaged according to similar material and design, and shall be marked for the type valve for which it is intended.

5.2.2.2 Packing. Onboard repair parts shall be packed in accordance with 5.2.1.2.

5.2.2.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, supplier's name and destination.

5.3 Domestic shipment and storage or overseas shipment. The requirements, and levels of packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.2).

(5.3.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.3.1.1 Packaging.

5.3.1.1.1 Level A. The valves shall be packaged in accordance with method III of MIL-P-116. Unit and intermediate containers shall conform to PPP-B-566, PPP-B-676, or PPP-B-636. Quantity per unit and intermediate container shall be as specified by the procuring activity.

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5.3.1.1.2 Level C. Packaging shall be sufficient to afford adequate protection against physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.3.1.2 Packing.

5.3.1.2.1 Level A. The valves packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-585	Class 3
PPP-B-636	Type SF, weather resistant class

The gross weight of wood type boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the box specification. Closures shall be in accordance with the applicable box specification and the appendix thereto.

5.3.1.2.2 Level B. The valves packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-591	Class I
PPP-B-601	Domestic
PPP-B-621	Class 1
PPP-B-585	Class 1
PPP-B-636	Type SF, domestic class

The gross weight of wood type boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Closures shall be in accordance with the applicable box specification and the appendix thereto.

5.3.1.2.3 Level C. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.3.1.3 Marking. In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. Flush valves covered by this specification are intended primarily for Navy shipboard use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2, 3.2.2 and 3.2.3).
(c) Whether inlet adapter is required (see 3.2.1).
(d) Centerline dimension if other than as specified in 3.2.1.1.
(e) Packaging, packing and marking requirements other than those required by 5.2 (see 5.3).

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
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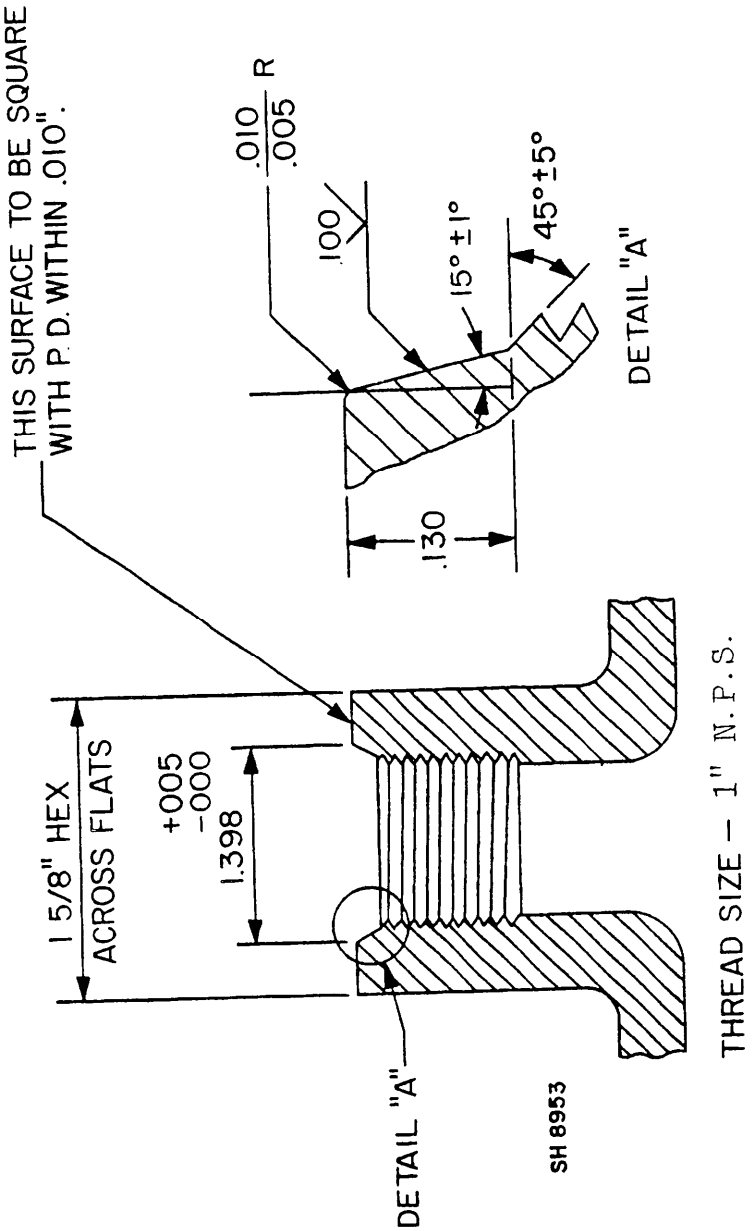
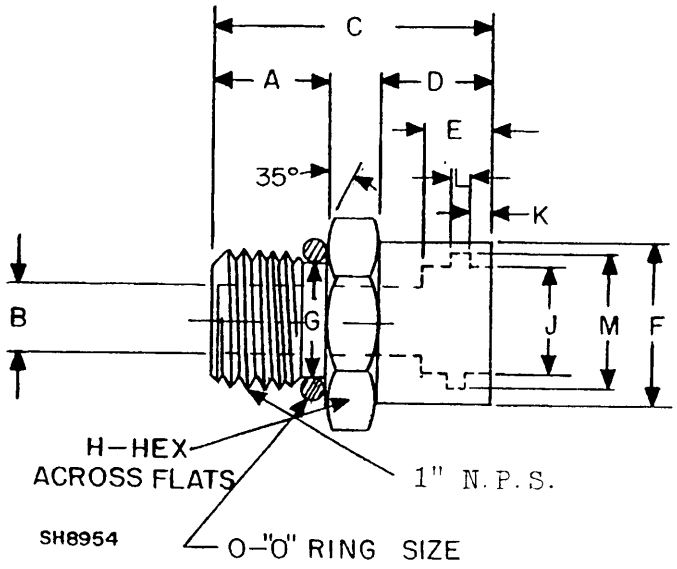
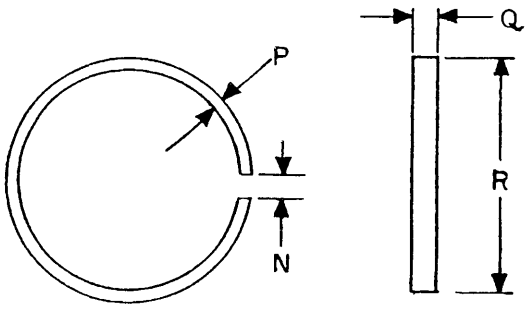


Figure 1 - Inlet to flow control stop valve.

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Dimensions in inches

Valve type	A	B	C	D	E	F	G	H	J		K	L		M		N		O		P	Q	R
									max.	min.		max.	min.	max.	min.	max.	min.	i.d.	o.d.	±.003	±.003	
I	0.594	1/0.875	2.000	0.760	0.438	1.535	1.195	1-5/8	1.318	1.315	0.156	0.140	0.130	1.444	1.424	0.135	0.073	1.171	0.116	0.050	0.125	1.444
II	.594	.697	1.840	.600	.375	1.020	1.195	1-5/8	0.843	0.840	.141	.093	.083	0.949	0.929	.130	0.073	1.171	0.116	.040	.078	0.949

1/ Type I only. Fair out 30 degrees to 1.171 at point of change from B to J.

Figure 2 - Inlet adapter for flow control stop valve.

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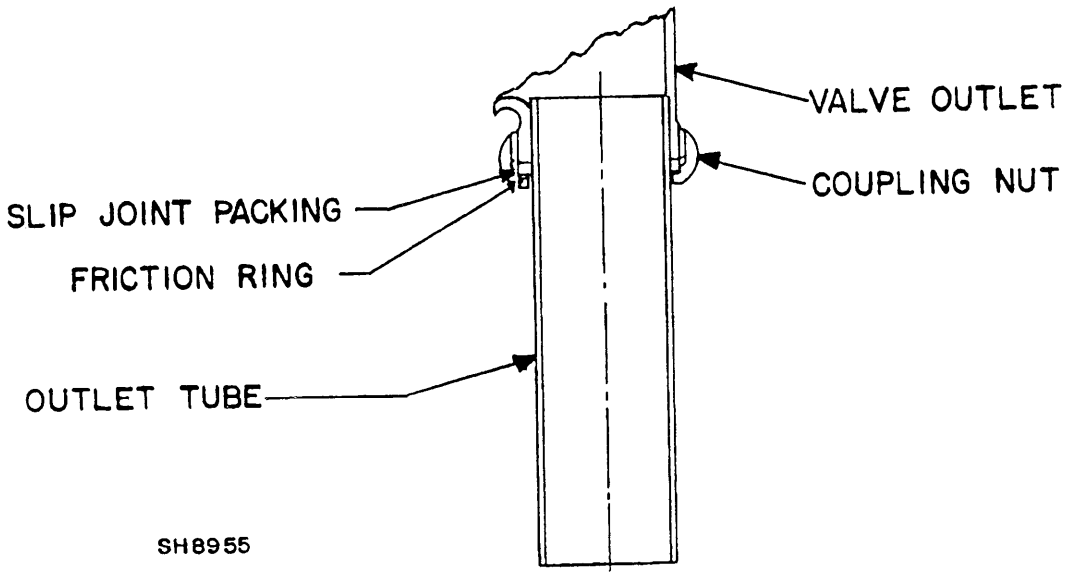


Figure 3 - Outlet connection of type I flush valve for shock mounted installations.

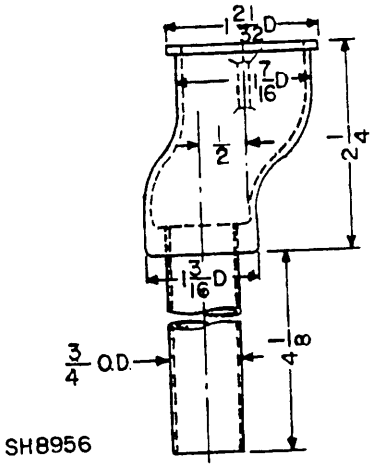


Figure 4 - Outlet connection of type II flush valve for shock mounted installation.

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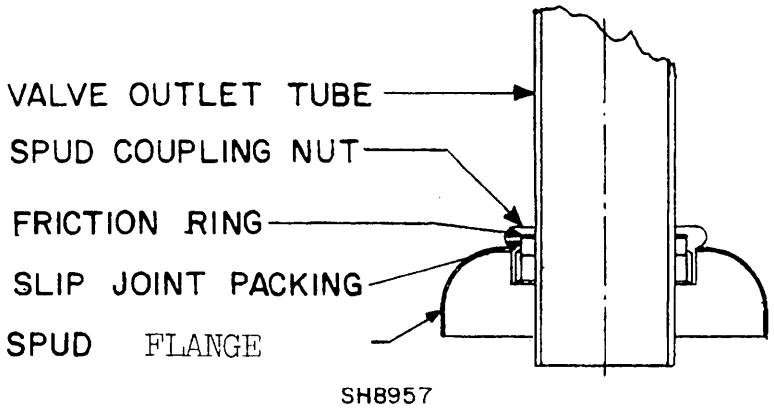


Figure 5. Connection of type I flush valve to rigid mounted installation.

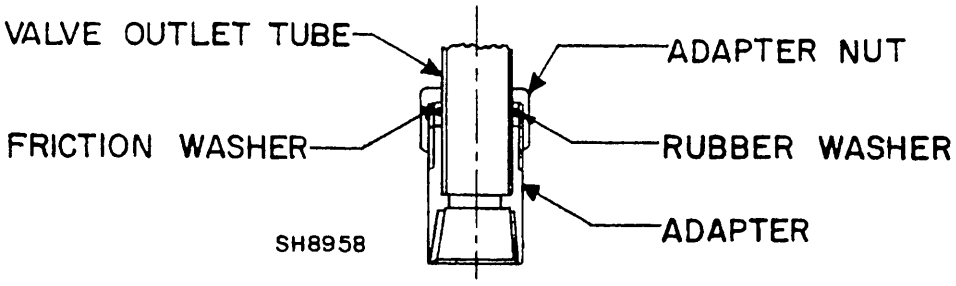


Figure 6 - Connection of type II flush valve to rigid mounted installation.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4963, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 12-66)

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